CCCA Meta-Atlas Cell-State Analysis

Li et al. 2019 · Melanoma (MARS-seq) · 1,082 malignant cells / 3 samples

GO / NO-GO GATES

Gate 1 GO

Mixed + continuous. Harmony theta-sweep read on the embedding + per-neighbourhood iLISI (1.00 -> 2.00 of 3 samples at theta=5); all 3 samples populate both density lobes (no patient archipelago). One connected component; lobes bridged. Sub-region biology settled: lobes are near-equal cycling (0.58/0.64) and no developmental module tracks proliferation (all |module vs G2M rho| weak) -> one developmental axis, whole manifold kept.

Gate 2 GO (non-branching)

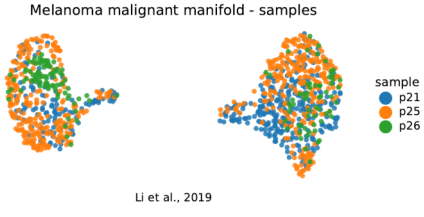
PPT tree over-branches within each lobe at every setting (13-20 forks) = noise-fitting, not a hub->two-arms bifurcation; the manifold is one elongated axis, fit by a fold-free curve (2 tips, 0 forks). Single Gast/pSom-high origin -> pSom-leaning axis -> inverse terminal Crest gradient (Crest-high decile pseudotime = 1.00). Crest is terminal, not on a Gast arm -> no inversion flag.

Gate 3 NA

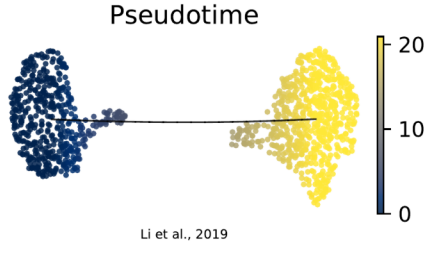
Branching-only. The manifold is non-branching (single continuous curve), so there is no bifurcation hub to test for Gast<->pSom cross-repression. Gate 3 not applicable.

STEP FIGURES

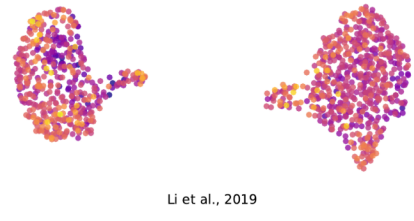
01 · Sample integration



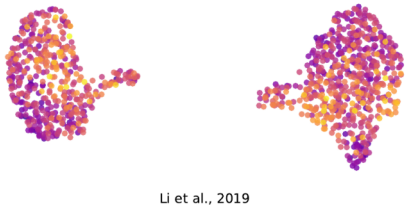
02 · Pseudotime + curve



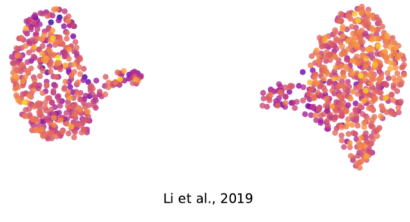
03 · Gastrulation



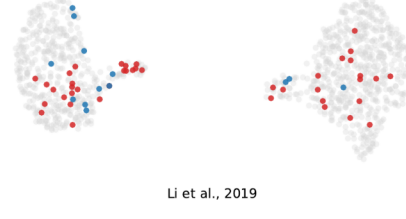
03 · Somitogenesis



03 · Neural crest



04 · EMT-TFs



Dataset	Melanoma, MARS-seq (Li et al. 2019). 1,160 malignant cells -> 1,082 after per-sample threshold (3 samples).
Key choices	Threshold = 56 (gap-finding: 154->21, 7.3x drop; derived). theta = 5 (near-max iLISI mixing; theta=10 fragments the layout). n_pcs = 20. Root = Gast/pSom-high curve tip (developmental origin; Crest is late) - not a proliferation state. Cell cycle scored (no publication phase calls).
Geometry	Non-branching, single continuous curve (Linear-pSom/Crest). Global means: Gast -0.015, pSom +0.016, Crest -0.002.
Hub / r	NA - no bifurcation hub (non-branching), so no Gast<->pSom cross-repression r to report.
EMT-TFs	FOXC1 (early / root) -> PRRX1 (late / terminus), detection-gated on this dataset.

VERDICT Pattern reproduces as the non-branching case: one embryonic developmental axis from a Gast/pSom-high origin to a Crest-high terminus, sample-integrated and continuous. G1 GO / G2 GO / G3 NA.